

1 S1 Appendix

2 Supporting information for: Source-only model selection and limits of fixed spectral-sharing
3 pipelines in cross-subject motor-imagery EEG decoding

4 **S1. Full internal model inventory and participant effects**

5 Table S1 lists all completed nine-person Q4–Q11 four-class conditions, including shallow
6 spectral/geometric controls and source-clean/source-normalized variants. The inventory is
7 descriptive; a higher target mean does not select a method for a confirmatory claim.
8 Accompanying CSV/JSON files provide participant values, fixed seeds, primary all-trial
9 populations and source file hashes. The complete inventory contains 61 Q4–Q14 condition
10 records, with repeated endpoints and the four single-person diagnostic cells identified. These
11 records are neither independent replications nor a cumulative fit count. Robustness conditions
12 appear in Table 5, source-composition means in the complete CSV, and the corresponding
13 statistical families in Table S2.
14 Table S1. Completed nine-person Q4–Q11 four-class conditions.

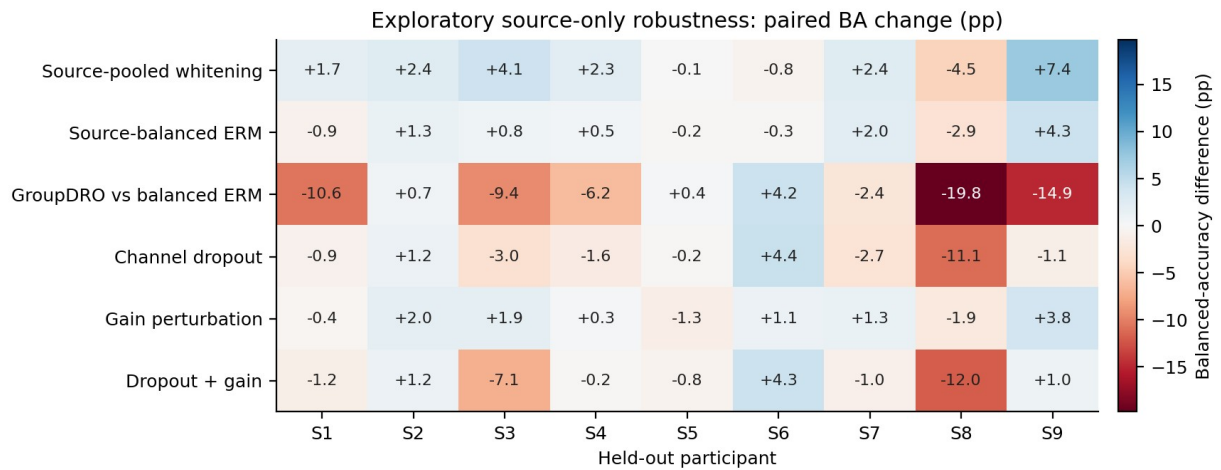
Archived condition	Mean BA (%)	Subject SD (pp)
Q4-E001/BroadCSP LDA	39.04	14.18
Q4-E001/BroadCSP SVM	37.50	12.66
Q4-E001/FBCSP LDA	38.12	7.69
Q4-E001/FBCSP MI8 LDA	31.37	4.26
Q4-A001/FBCSP MI16 LDA	34.65	8.45
Q4-A001/FBCSP MI32 LDA	36.75	8.86
Q4-A001/FBCSP MI72 LDA	38.12	7.69
Q4-A001/FBCSP MI8 LDA	31.37	4.26
Q5-E001	33.72	11.58
Q6-E001	37.93	15.68

Archived condition	Mean BA (%)	Subject SD (pp)
Q8-E001	42.67	16.04
Q9-A001/PSD44 LDA	36.28	8.17
Q9-A001/PSD44 LINEAR SVM	35.78	8.39
Q10-A001/BROAD LOGE LOGREG	35.32	10.74
Q10-A001/BROAD LOGE MDM	29.49	8.73
Q10-A001/MUBETA LOGE LDA	36.00	10.75
Q10-A001/MUBETA LOGE SVM	35.47	8.83
Q9-E001/BETA 13 30	31.53	10.05
Q9-E001/MID 8 30	44.55	17.17
Q9-E001/MU 8 13	43.30	16.22
Q9-E001/MU BETA SHARED	42.19	14.10
Q9-E002/MID 8 30 Q8 EPOCHS	42.73	15.73
Q9-E002/MU BETA SHARED Q8 EPOCHS	43.50	15.82
Q9-E004/MU BETA SHARED SOURCE CLEAN	42.11	15.23
Q9-E005/MU BETA SHARED SOURCE NORM	41.53	13.88
Q10-E001/MU BETA CSP8 EEGNET	41.48	16.04
Q10-E001/MU BETA PCA8 EEGNET	42.06	15.51
Q11-E001/BROAD CAPACITY MATCHED	42.19	14.05
Q11-E001/FOUR BAND SHARED	39.80	11.58

Archived condition	Mean BA (%)	Subject SD (pp)
Q11-E001/TWO BAND EARLY STACK	42.66	16.23
Q11-E001/TWO BAND INDEPENDENT	42.99	17.38

15 Neural seeds are averaged within person; shallow arms evaluate one model per fold. Q4-A001
16 reuses fitted CSP feature caches but refits scaler/LDA. Its k8/k72 predictions reproduce earlier Q4
17 endpoints and add no independent outcomes. Condition IDs record protocol order rather than a
18 leaderboard ranking. The four S3 diagnostic cells appear separately in Fig 2C, outside the nine-
19 person rows.

20 S2. Robustness heterogeneity



21
22 Figure S1. Participant-level robustness differences for all nine people in fixed order. Colours and
23 annotations show balanced-accuracy percentage points. GroupDRO uses balanced ERM as its
24 control; all other rows use the historical Q8 mean-rank baseline and cross runtimes. The display is
25 descriptive, with no target-based exclusions or significance stars.

26 **S3. Duration, source-count, and source-session statistical families**

27 Table S2. Archived exploratory Q13/E006 paired contrasts.

Contrast	Δ (pp)	95% CI (pp)	Holm p
Broad k2 minus k8	-11.03	[-18.80, -3.56]	0.1641
Broad k4 minus k8	-5.10	[-10.36, -0.38]	0.4336
Broad k6 minus k8	-0.83	[-3.67, +1.89]	0.5664
Shared k2 minus k8	-9.23	[-15.41, -3.15]	0.1641
Shared k4 minus k8	-5.09	[-9.28, -1.31]	0.2344
Shared k6 minus k8	-1.49	[-3.41, +0.42]	0.4336
Broad 1test minus 0train	-0.84	[-1.89, +0.19]	0.3516
Shared 1test minus 0train	+0.08	[-1.14, +1.18]	0.8516
Fixed20 minus Historical CE	+10.27	[+0.96, +21.62]	0.1562
Shared FIXED20 minus Shared RAW CE	+3.91	[+0.44, +7.62]	0.1562
Matched CE minus Fixed20	-10.06	[-21.53, -0.68]	0.1250

28 Each row summarizes nine people after averaging fixed seeds and, where applicable, source
29 subsets within person. The historical CE contrast crosses runtimes; the separately declared
30 matched-CE-minus-fixed20 contrast controls runtime. Intervals are unadjusted, and p-values are
31 corrected within each declared family. Signs follow the archived contrasts; Fig 3B reverses the
32 matched contrast for readability.

33 **S4. Audit and reporting boundaries**

34 The manuscript-number audit reaggregates stored predictions independently and records the CSV
35 hashes used. Historical scientific replay counts are reported only when supported by a passing
36 scientific receipt. The Q9 top-level batch report documents orchestration, and the versioned

external portability validator adds no inference. CRLF-to-LF matches are identified explicitly. The paper bundle contains the source-snapshot manifest, review input hashes, independent numerical audit, generated tables, figure sources and rendering checks. These records document existing analyses; they do not authorize training or establish an external outcome-selected method freeze. For completed Q15, the final scientific validator binds 221 artifacts and verified result publication binds 223 artifacts at the committed snapshot. The manuscript independently checks these bindings, 52 source-artifact bindings and 10 frozen checkpoint/transform bindings; freeze and manifest documents are checked separately. Status files can change after the result commit, so each hash is checked at the commit named by its receipt. An intermediate receipt marked pending does not override the final passing validator.

The original mean-rank follow-up reports a bootstrap interval of $[+1.048, +19.959]$ pp (20,000 draws, seed 20260923); independent reanalysis with 200,000 draws and seed 20261002 gives $[+1.067, +19.952]$ pp. The main text retains the original interval. The saved nine-person sign sensitivity includes zero changes in its denominator; the conventional tie-excluding sensitivity is reported separately. Both implementation and resampling differences are disclosed to explain the small numerical discrepancy.

The external primary sign test retains the archived exact-zero rule. S10 has a saved shared-minus-broad difference of approximately -1.11×10^{-16} and counts as negative: 63 positive, 43 negative and three zero differences, $p = 0.06446$. With a numerical tolerance of 10^{-14} , that difference becomes a tie, giving 63 positive, 42 negative, four ties and $p = 0.05044$. The primary rule remains unchanged. Neither calculation establishes superiority at the conventional 0.05 threshold.

S5. Earlier binary calibration, artifact, and EOG checks

The earlier pipeline checks concern left/right imagery and are reported separately from the four-class analysis. P2-E001 contains 2,346 expert-clean trials. Within-session leave-run-out and cross-session models use labelled trials from the evaluated person; cross-subject LOSO excludes both

63 target sessions. Within-session/LOSO evaluate all 2,346 trials, whereas cross-session evaluates
64 1,183 later-session trials. P2-SMOKE-S1B is a single-person technical check, not a population
65 replication. P2-E002-ALLTRIALS includes 2,592 trials in both source and target populations.
66 Some retained metadata still carries an earlier clean-population name; interpretation follows the
67 actual trial populations.

68 P3-E001 compares CSP4, Welch PSD88 and their 92-feature fusion with source-all and source-
69 expert-clean training. Its primary endpoint uses the same 2,346 expert-clean target trials; all 2,592
70 trials and the 246 flagged trials are secondary strata. P4-E001B is the successful EOG-processing
71 retry. Regression coefficients fitted on clean source data are applied to three synchronous target
72 EOG channels, without target parameter fitting. The method therefore requires these additional
73 sensors. Clean-target balanced accuracy was 62.57% versus 62.85% without correction,
74 establishing neither a decoding benefit nor selective ocular-artifact removal. The failed original P4
75 attempt remains in the technical record.

76 Table S3. Earlier binary QC primary endpoints; all nine participants.

Stage / model	Split / source policy	Target trials	BA (%)
P2-E001 / CSP+LDA	cross session; expert clean only	1183	76.06
P2-E001 / CSP+linearSVM	cross session; expert clean only	1183	75.05
P2-E001 / CSP+LDA	cross subject; expert clean only	2346	62.76
P2-E001 / CSP+linearSVM	cross subject; expert clean only	2346	62.09
P2-E001 / CSP+LDA	within session; expert clean only	2346	78.06
P2-E001 / CSP+linearSVM	within session; expert clean only	2346	78.21

Stage / model	Split / source policy	Target trials	BA (%)
P2-E002-ALLTRIALS / CSP+LDA	cross subject; all trials	2592	61.50
P2-E002-ALLTRIALS / CSP+linearSVM	cross subject; all trials	2592	60.57
P3-E001 / CSP4+PSD88 + LDA	cross subject; all trials	2346	64.10
P3-E001 / CSP4 + LDA	cross subject; all trials	2346	61.92
P3-E001 / PSD88 + LDA	cross subject; all trials	2346	58.55
P3-E001 / CSP4+PSD88 + LDA	cross subject; expert clean only	2346	64.63
P3-E001 / CSP4 + LDA	cross subject; expert clean only	2346	62.85
P3-E001 / PSD88 + LDA	cross subject; expert clean only	2346	58.19
P4-E001B / EOG regression	cross subject; source train fitted EOG regression	2346	62.57
P4-E001B / Uncorrected CSP4 + LDA	cross subject; uncorrected	2346	62.85

77 All rows are binary, with 50% chance. Within-session and cross-session rows use target-person
78 labelled calibration. P3/P4 rows use expert-clean targets; P2 all-trial rows use all trials. CSV
79 supplements retain all/flagged secondary endpoints, target-information notes, and the single-
80 person smoke. These modes and populations are not pooled into a calibration-free score.

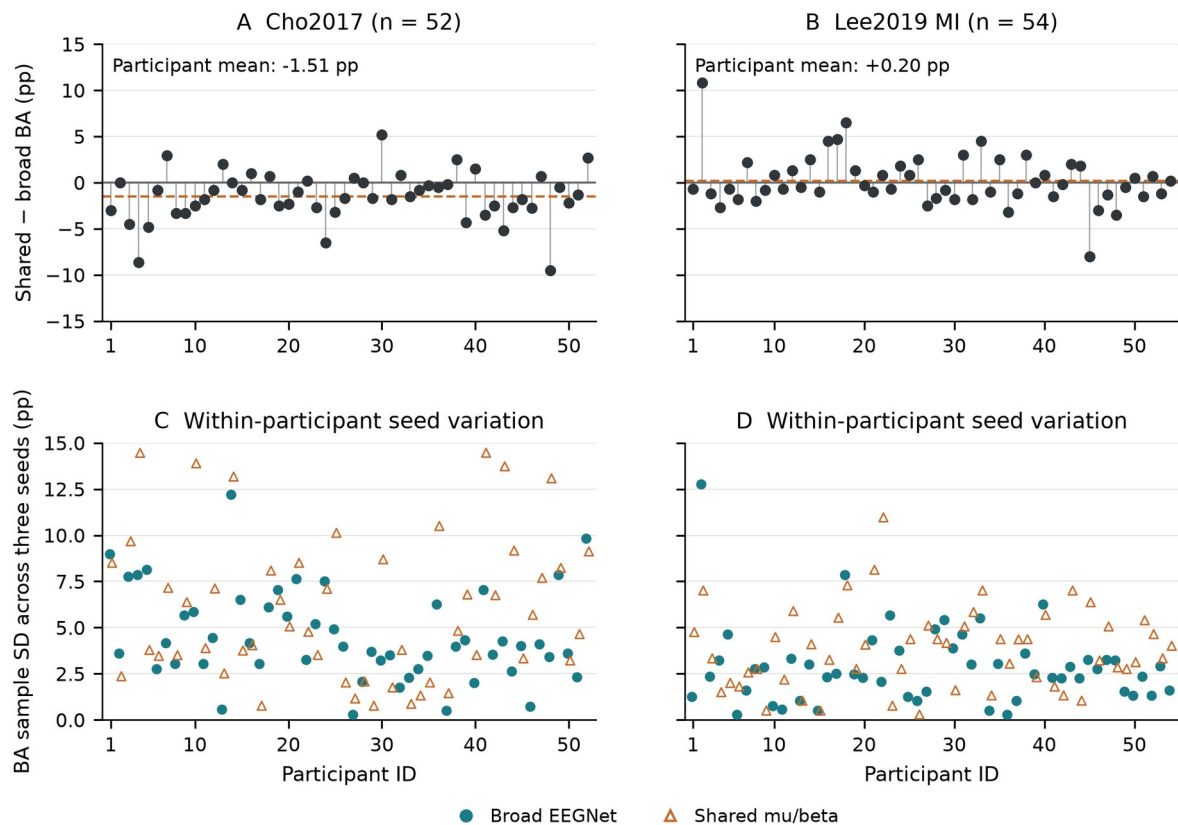
81 **S6. Selection-curve analyses and technical history**

82 Q8-A001/A002/A003 inspect archived source-validation curves without further fitting. Leave-one-
83 inner-fold-out sensitivity and six candidate aggregation rules generate hypotheses; they are not six
84 target-evaluated decoder conditions. P1-E001 is a single-subject acquisition audit. P2-SMOKE-S1

85 and the original P4-E001 retain failure receipts followed by successful retries. Q13-E002/E003 and
 86 Q14-E003 were not run and have no performance values. The partial 87-person external attempt is
 87 not an additional cohort alongside the completed 109-person evaluation. The completed-
 88 experiment inventory records each analysis's evidential role, reuse, validation scope and technical
 89 failures.

90 **S7. External adapter, freeze chronology, and cohort-level sensitivity**

91 The companion supplementary_methods.md specifies the 21-channel montage, six-second native
 92 context, deterministic cue indexing, native-rate filters, two-second crop, Helmert coordinates,
 93 provider label maps, raw audits, source validation groups, 15 original fits and source/inference
 94 freeze chronology. Cho2017 retains labelled imagery: 49 people contribute 200 trials each, and S7,
 95 S9, S46 contribute 240. Lee2019 includes offline-training runs only, with 100 trials per session.
 96 All primary participants are retained, and Q15 interpretations retain the stated calibration
 97 limitations.



All 106 participants retained. Sessions are pooled within participant; seeds are not independent participants.

99 Figure S2. Participant-level Q15 transfer heterogeneity. Saved seed-averaged shared-minus-broad
100 effects and within-person three-seed sample standard deviations (106 paired effects and 212
101 dispersion points; CSP omitted) are displayed separately for 52 Cho and 54 Lee people. Participant
102 order is descriptive and does not define subgroups or target selection. Repeated seeds and the two
103 Lee sessions are not additional inference units.

104 Shared input had greater descriptive seed dispersion than broad input: mean within-person seed SD
105 was 6.014 versus 4.550 pp on Cho, and 3.798 versus 2.887 pp on Lee. These SDs summarize three
106 fixed source-model seeds, not a population distribution of training randomness, and are not
107 additional confirmatory hypotheses. Person/seed metrics and class recalls are supplied in
108 q15_external_subjects.csv, q15_seed_metrics.csv and q15_model_summary.csv.

109 **S8. Reproducible manuscript materials**

110 The submission materials contain the manuscript, supporting information, standalone figures,
111 frozen derived data and validation evidence. All manuscript-preparation model fits and
112 checkpoint-inference counts are zero. Operational cloud logs are not scientific endpoints. Author
113 declarations and final submission checks are maintained separately from the scientific results.

114 **S9. Descriptive BNCI-only Physiological Characterization**

115 The separate Q16 methods specify native raw-file/hash and event checks, fixed Welch parameters,
116 trial-level dB aggregation, hand descriptors, artifact/session strata and all-nine-person associations
117 with saved Q14-E001 binary LOSO scores. The specification follows known decoder outcomes. It
118 adds no external physiology, p-values, subgroup selection, decoder fits or decoder inference. The
119 public parameter freeze is 050e01b028aaab8e3d745934b13b2d17e9bb0a7a. Independent
120 validation passed for the central-channel raw power replays, all saved aggregation tables and all
121 six signed correlations. The bundle retains the freeze/readback, raw and execution receipts,
122 summary, validation, person/session/artifact tables and figure input/export hashes.

123 Table S4. All six descriptive physiology–balanced-accuracy associations.

Saved binary LOSO model	Band	Spearman rho	People	Context
Broad EEGNet	Mu	−0.150	9	Designated descriptive endpoint
Broad EEGNet	Beta	+0.600	9	Designated descriptive endpoint
Shared mu/beta EEGNet	Mu	−0.083	9	Context
Shared mu/beta EEGNet	Beta	+0.483	9	Context
CSP4 + LDA	Mu	+0.067	9	Context
CSP4 + LDA	Beta	+0.317	9	Context

124 Spearman rho is the Pearson correlation of average ranks. Each person contributes one equal-
125 session signed C3/C4 hand descriptor and one saved Q14-E001 binary LOSO balanced-accuracy
126 score: a three-seed mean for neural models and one CSP fit. All nine people are retained. Positive
127 rho associates higher balanced accuracy with a larger signed descriptor, which is less relatively
128 suppressive when negative. Coefficients are rounded to three decimals here and retained at full
129 precision in the data tables. These post-decoder-outcome associations have no p-values,
130 confirmatory claims or external physiological test.

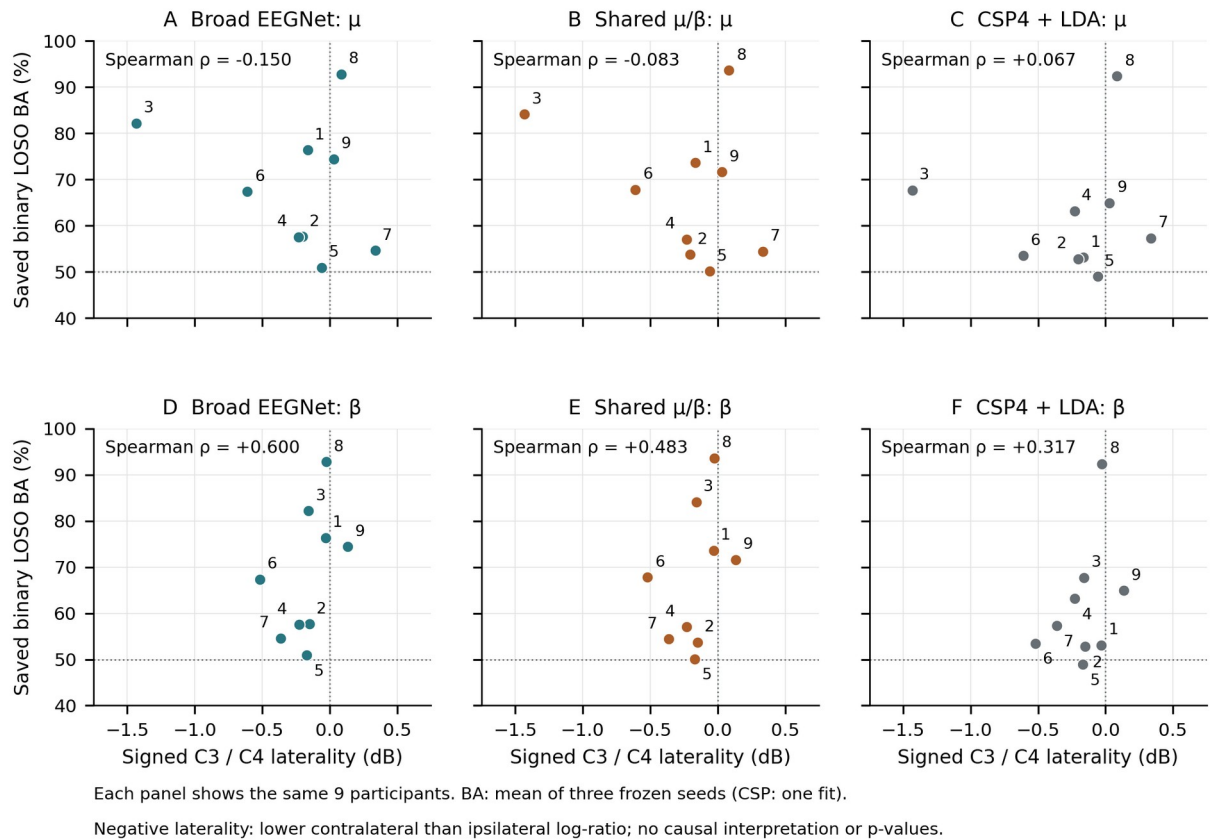


Figure S3. All six descriptive BNCI associations. Every model/band panel shows the same nine people, labelled by person ID. The horizontal axis is signed C3/C4 hand laterality; the vertical axis is saved Q14-E001 binary LOSO balanced accuracy. Neural scores average three seeds within person; CSP uses one frozen fit. The physiology window [0.5,2.5) s differs from the decoder window [0.5,3.5) s. All points are retained, with no fitted lines, p-values, subgroup thresholds or decoder updates. Positive beta correlations associate higher balanced accuracy with less-negative laterality. These post-outcome descriptions establish neither mechanism nor causality and do not validate a biomarker.

Detailed reproducible methods

Supplementary methods

Companion to *Source-only model selection and limits of fixed spectral-sharing pipelines in cross-subject motor-imagery EEG decoding*. Revision dated 7 October 2026. This document describes

completed decoder analyses including Q15 and the completed, independently checked descriptive
BNCI-only Q16 analysis. Decoder source snapshot:
7af1a137e2676a018e1e880ab076de6cae4ce30b. The earlier manuscript inventory is retained as
historical evidence rather than treated as the final dataset definition.

S1. Study sequence and units of evidence

Source-only decoding procedures were developed on nine BNCI2014_001 participants and
evaluated as frozen binary pipelines on separate external acquisition cohorts held out from fitting.
Participant identity nonoverlap across providers was not independently established. The four-class
development task, Q14 binary PhysioNet evaluation and Q15 binary Cho2017/Lee2019_MI
evaluation use different input contracts, so balanced accuracies are reported separately. No pooled
accuracy across tasks or combined external participant-level significance test was specified.
Strict zero-calibration is defined within each evaluation fit: the current target person's recordings
and labels do not estimate or select weights, BatchNorm state, shallow coefficients, preprocessing
coefficients, training duration, seeds, thresholds or model choice for that fit. Source-fitted
transformations are applied unchanged. Fixed filtering, channel selection, resampling, CAR and
analytical Helmert coordinates compute functions of target signals without fitting a target-
dependent decoder. Event and class metadata implement declared eligibility and canonical label
maps; labels score predictions. Separate Q16 summaries read native BNCI signals and labels
without changing any decoder. This information contract does not imply that the development
programme was designed without prior BNCI outcomes, that target labels were never accessed, or
that physical voltage calibration is known. Q4–Q13 are development and sensitivity analyses on
repeatedly used participants. Q15's pre-fit freeze was committed on 4 October 2026 at 01:33:03
UTC (fc0e7d54006076bec7701064e1045034c02d06a4), before source completion at 01:52:42
UTC. Its separate inference/statistics freeze was committed on 5 October at 09:31:05 UTC
(2ad479f5bccb4f92cd6c77bc78a8ba9b620604c1), before target inference completion at 10:57:19

169 UTC. This chronology is documented in the repository, rather than in an independently registered
170 clinical or trial registry.

171 The inventory distinguishes evaluated conditions, fitted models, reused endpoints, metadata audits
172 and unsuccessful or unrun branches. Its 61 historical Q4–Q14 condition rows and 34 earlier
173 quality-control rows are not 95 independent replications. Q15 adds six external model-by-cohort
174 endpoints. Three source-model records document the separate source fit/freeze stage and carry no
175 external performance claim.

176 **S2. Historical BNCI development and quality-control analyses**

177 BNCI2014_001 contains 5,184 trials from nine people, two sessions each, and four classes. Each
178 participant contributes 576 trials; the local complete metadata audit records 488 expert artifact-
179 flagged trials. The primary four-class analyses include all trials and consequently do not use the
180 original competition's artifact-free evaluation population. Within-session and cross-session P2
181 checks use labelled calibration from the evaluated participant, whereas cross-subject analyses
182 exclude both target sessions. These calibration modes remain separate in the inventory.

183 The primary historical neural pipeline filters native 250-Hz recording runs with a fourth-order
184 Butterworth prototype in zero-phase mode. Its half-open epoch is [2.5,5.5) s after stored trial start,
185 or [0.5,3.5) s after the imagery cue: 22 EEG channels and 750 samples. There is no baseline
186 subtraction, additional reference, or additional notch in that primary pipeline. MNE-loaded volt
187 values are converted to numerical microvolt inputs. Q5/Q8 use 4–40 Hz; the restricted broad Q9
188 arm uses 8–30 Hz. The shared model uses fixed 8–13-Hz and 13–30-Hz views.

189 Each outer LOSO fold excludes both sessions of one person. The remaining eight sorted source
190 IDs form four consecutive two-person validation groups. Inner fits train on six people and validate
191 on two; final fits train on eight and evaluate the held-out person. Q5 chooses the earliest minimum
192 of equal-fold mean validation cross-entropy. Q8 chooses the earliest minimum of the equal-fold
193 mean within-fold cross-entropy rank, with average ranks for ties, across epochs 1–40. It reuses 36
194 Q5 inner trajectories and performs 27 final fits. Representation-specific conditions generally

195 recompute source-only trajectories; fixed-duration controls retain their documented epoch
196 schedule.

197 Three final neural seeds, 20260924, 20260925, and 20260926, are averaged within person. Source-
198 subset repetitions are also aggregated within target person before population summaries. Neither
199 seeds nor overlapping source subsets are treated as independent participants. Detailed historical
200 controls and their limits are preserved in `supplementary_inventory.md` and the locally retained
201 historical method audit at `evidence/previous_draft/methods_and_references.md`.

202 The Q5 raw-versus-Q6 source-normalized comparison changes both the input scaler and two
203 selected schedules: S3 changes from two to 16 epochs and S5 from 13 to 12. The population
204 normalization delta therefore does not isolate normalization alone. Q7's raw/normalized $\times$ two/16-
205 epoch four-cell diagnostic fixes duration within each pair for S3, but supplies only a one-person
206 diagnostic. Earlier within-person P2 analyses use labelled target-person calibration and remain
207 outside strict zero-calibration endpoints.

208 The P3 clean/all/flagged strata overlap and must not be added as independent samples. P4-E001B
209 is the successful EOG-assisted retry: regression coefficients are estimated from clean source data
210 and applied with synchronous target EOG. A lower EEG–EOG correlation is not proof of selective
211 ocular artifact removal. The failed P4-E001 channel-selection attempt is retained separately. Q9
212 orchestration success supports execution bookkeeping; it does not by itself establish raw-to-
213 checkpoint replay. Later Q12/Q13 replay receipts have experiment-specific scopes.

214 **S3. Shared spectral representation and comparator definitions**

215 The shared μ/β pipeline supplies two fixed filtered views to the same EEGNet and averages their
216 class-logit vectors. Training cross-entropy is computed from the averaged logits; inference applies
217 softmax to that average. Both views form one two-view mini-batch, sharing BatchNorm
218 parameters and running statistics as well as convolutional weights. Dropout is applied per view
219 during training. This imposes an input and parameter-sharing constraint. It does not learn attention,
220 train two independent networks or average probabilities.

221 The configured EEGNet uses F1=8, D=2, F2=16, a 64-sample temporal kernel, and dropout 0.25.
222 Training uses Adam, learning rate 0.001, batch size 64, weight decay 0, and cross-entropy without
223 a learning-rate scheduler. Model evaluation and inference mode prevent target examples from
224 updating BatchNorm. Parameter counts depend on input duration and output classes; the historical
225 22×750 four-class model's 2,932 parameters must not be copied to a different Q15 architecture
226 without checking it.

227 Historical shallow comparators are implementation-specific. The Q4 filter-bank controls use OAS
228 regularization and shrinkage LDA or linear SVM. Q14/Q15 CSP4+LDA instead use unregularized
229 CSP, concatenated covariance estimation, log power, no trace normalization, full-rank coordinates,
230 mutual-information component order, source-fitted StandardScaler, and LDA with the SVD solver.
231 These comparators are not interchangeable. Q15's fixed Helmert coordinates, described below,
232 address the rank loss induced by its common-average reference.

233 **S4. Q14 external evaluation on PhysioNet**

234 Q14 uses separate binary BNCI source models trained on left/right imagery only: 2,592 trials, 288
235 per source participant. Native filtering and 750-sample historical epochs precede polyphase
236 resampling from 250 to 160 Hz, producing 22×480 inputs. All-nine-source duration selection uses
237 validation groups [1,2], [3,4], [5,6], and [7,8,9]. Their training complements contain seven, seven,
238 seven, and six people. Equal-fold weighting is retained despite unequal group sizes. Selected
239 durations are 18 epochs for broad EEGNet and 17 for shared μ/β . This source stage comprises
240 eight inner and six final neural fits, plus one CSP fit. The 18/17-epoch contrast compares complete
241 selected pipelines rather than isolating parameter sharing.

242 The completed external Q14-E002R2 evaluation uses all 109 PhysioNet participants and imagery
243 runs 4, 8, and 12 only. T1 maps to left-hand imagery and T2 to right-hand imagery; rest, executed
244 movements, and bilateral tasks are excluded. It comprises 327 EDF files and 4,918 distinct trials.
245 Two neural models × three seeds plus one deterministic CSP model produce 34,426 prediction
246 rows. Those rows do not represent 34,426 independent observations.

247 The metadata-driven R2 amendment handles three native 128-Hz participants (88, 92, and 100);
248 the remaining 106 are native 160 Hz. Native filtering/epoching precedes deterministic resampling
249 to 480 samples. The full 109-person endpoint supersedes the partial 87-person R1 attempt. A later
250 portable validator replays archived event identities, saved probabilities, and statistics in its
251 documented validation runtime; that scope should not be described as the Q15 full raw/checkpoint
252 replay. No target fit, target normalization, or target-based duration choice occurs.

253 **S5. Q15 source and target cohort definitions**

254 Q15 is an explicitly amended 21-channel external benchmark, not an unchanged rerun of the Q14
255 preprocessing. Q15-E005 performs source-only fitting on the nine BNCI participants, preserving
256 the binary population of 2,592 trials across both sessions. The new input is 21×320 at 160 Hz.
257 Raw-file audits and a committed pre-fit freeze precede every source fit. The source validator
258 verifies eight inner plus six final neural fits and one shallow fit. Its selected durations are 14
259 epochs for broad EEGNet and 19 epochs for shared μ/β . These are distinct from the Q14 18/17
260 durations. The Q15 broad/shared contrast includes both the input design and unequal source-
261 selected schedules, so it is a complete-pipeline comparison rather than a causal sharing effect.
262 Cho2017 includes 52 people and 52 raw MAT files. The retained imagery_left and imagery_right
263 signals comprise 10,520 trials. Most participants have 100 trials per class; participants 7, 9, and 46
264 have 120 per class. Imagery event indices use the audited zero-based nonzero event-array samples,
265 with no added timing shift. Data concatenated by class are split into independently retained trial
266 contexts. The study does not reconstruct original physical run identities; retained_labeled_MI is an
267 analysis label, not a verified acquisition run identifier.
268 Lee2019_MI includes 54 people, two sessions each, and 108 MAT files. Only EEG_MI_train is
269 evaluated: 100 labelled trials per file and 10,800 trials overall. EEG_MI_test and other tasks are
270 excluded. Cue indices use stored MATLAB t minus one. The native class convention is mapped as
271 1 → canonical right=2 and 2 → canonical left=1. The stored segmented smt alignment differs from

272 this explicit cue convention and is not silently substituted for it. Both sessions are pooled within
273 person for the primary metric.

274 The retained source/target channel order is Fz, FC3, FC1, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6,
275 CP3, CP1, CPz, CP2, CP4, P1, Pz, P2, and POz. FCz from the BNCI 22-channel montage is
276 omitted by the frozen common-channel rule. No channel interpolation, imputation, or target-fitted
277 montage correction is used.

278 **S6. Q15 numerical preprocessing contract**

279 For every dataset, the adapter retains a complete half-open six-second native context $[-1.5, 4.5)$ s
280 relative to the audited cue. A missing or truncated context is an error; it is not shortened or padded
281 into an accepted trial. Channel selection precedes an instantaneous arithmetic common-average
282 reference across the 21 selected channels. Referencing and filtering operate in float64. Each
283 retained trial context is processed independently, avoiding filtering across artificial class-
284 concatenation boundaries in Cho2017.

285 Three native-rate filters are fixed: broad 8–30 Hz, μ 8–13 Hz, and β 13–30 Hz.

286 `scipy.signal.butter(4, ..., btype="bandpass", output="sos")` supplies a fourth-order low-pass
287 prototype, yielding an eighth-order bandpass represented by four second-order sections. `sosfiltfilt`
288 applies forward/backward filtering with odd extension and `padlen=27` native samples. The
289 coefficient tables for 250, 512, and 1,000 Hz and the NumPy/SciPy versions are embedded in the
290 freeze. Zero-phase filtering is offline and noncausal; these results do not establish online causal
291 performance.

292 The complete filtered context is resampled with `scipy.signal.resample_poly`, a Kaiser window
293 parameter of 5.0, constant boundary extension, and `cval=0`. Ratios are 16/25 for BNCI at 250 Hz,
294 5/16 for Cho2017 at 512 Hz, and 4/25 for Lee2019_MI at 1,000 Hz. The common context contains
295 exactly 960 samples. A half-open slice `[320:640]` yields the $[0.5, 2.5)$ -s cue-relative window and
296 320 model samples. Final inputs are contiguous float32. No baseline correction, artifact rejection,

target moment fitting, target covariance alignment, or target-based duration/seed selection is introduced.

The 21-channel common-average signal has at most 20 spatial degrees of freedom. Before Q15 CSP, the complete referenced subspace is represented by the fixed 20×21 orthonormal Helmert matrix `scipy.linalg.helmert(21, full=False)`. This basis depends only on channel count; it is not estimated from either dataset. The source-only CSP pipeline then retains four components. The basis is not target adaptation and is not a learned dimension-selection result.

The declared convention is native numerical values interpreted as microvolts. Provider file identities, hashes, shapes, channels, sampling grids, cues and label conventions were audited; physical export voltage calibration was not independently established. The original Cho hardware reference and cue latency also remain unverified. Common-average referencing defines numerical coordinates without authenticating the original hardware reference. Q15 is therefore reported as `completed_with_calibration_limitations`: an adapter benchmark under a stated input convention. Amplitudes and between-dataset differences do not establish voltage equivalence or a controlled acquisition mechanism.

S7. External inference, statistics, and validation

Six final neural checkpoints and one deterministic shallow model are frozen before target inference. Each target trial has seven prediction rows: two neural models with three seeds each and one shallow model. Cho2017 contains 73,640 rows for 10,520 distinct trials; Lee2019_MI contains 75,600 rows for 10,800 trials. Original source fits total 15; migration performs zero new source fits and zero target fits. Reuse is established by the checkpoint/artifact bindings and preserved source validation, rather than inferred from a launcher receipt.

For person i and neural seed s , balanced accuracy is the mean of left-class and right-class recall. Lee sessions are pooled within person before scoring; session scores remain descriptive. The three seed-specific balanced accuracies are then averaged within person, and cohort balanced accuracy averages those person values equally despite Cho's unequal trial counts. The primary paired

323 contrast is shared μ/β minus broad EEGNet within person. CSP4+LDA provides context and is
324 outside the predeclared primary family.

325 For each cohort, the 95% interval is the percentile interval from 20,000 bootstrap resamples of
326 whole people, preserving the pairing, with seed 20260924. The two-sided sign-flip Monte Carlo
327 test independently assigns a random ± 1 sign to each person's paired difference, uses 20,000 draws
328 and seed 20261003, and applies the plus-one p-value correction. This tests a sign-exchangeability
329 null; it does not turn correlated training seeds or individual trials into independent observations.

330 Holm adjustment covers exactly the two predeclared cohort contrasts. The procedure is frozen
331 before target predictions, and no pooled-cohort effect, post-hoc target tuning, equivalence margin,
332 or target-based model selection is claimed.

333 Cho2017 mean balanced accuracy is 59.8237% for broad EEGNet, 58.3104% for shared μ/β , and
334 51.9904% for CSP4+LDA. The shared-minus-broad contrast is -1.5134 percentage points, 95%
335 CI $[-2.2490, -0.8045]$, with Holm-adjusted $p=0.000099995$. Lee2019_MI means are 65.5123%,
336 65.7160%, and 52.7685%, respectively; its contrast is $+0.2037$ percentage points, 95% CI
337 $[-0.5154, +0.9691]$, Holm-adjusted $p=0.60367$. The Cho result opposes a general shared-band
338 superiority claim. The Lee result provides no clear paired advantage; absence of a significant
339 difference does not establish equivalence.

340 The final independent computational Q15 validator replays the raw metadata audit, raw-to-epoch
341 transformation, frozen checkpoint predictions, coverage, labels, and statistical calculations. It
342 verifies unchanged neural model state and a shallow-model digest before/after inference.

343 Probability replay uses $\text{atol}=10^{-7}$ and $\text{rtol}=10^{-6}$, with exactly matching argmax predictions. The
344 passing results/Q15-EXTERNAL/validation_report.json supersedes the temporal inference-
345 completion status as completion evidence. Intermediate statistics.json, holm_two_cohorts.json, and
346 completion_receipt.json retain their pre-validation pending labels by design; their numeric/hash
347 bindings are checked by the final validator. This replay verifies computational consistency and
348 state immutability; it is not an additional cohort or independent laboratory replication.

349 **S8. Runtime and provenance**

350 Source fitting completed on 4 October 2026 in the origin job 20261004T005335Z-9b3bce30277a.
351 Q15 continuation completed and was validated in the migration job 20261005T050511Z-
352 migration-from-r2-b82ad79b. The scientific revision is
353 271af288a2f3863430ab80e3145c2dee9bd5571d; the source/artifact continuation base is
354 782d2d0070a50c37d13c8e9f1cab3b3b81bac4fc. Final scientific-results publication was read back
355 at bc48b257eb44f412ad069f50d0f1a72a33c3c520; the inspected results branch snapshot is
356 7af1a137e2676a018e1e880ab076de6cae4ce30b.

357 The source runtime records Python 3.12.3, Linux/glibc 2.39, NVIDIA RTX 4000 Ada Generation,
358 PyTorch 2.8.0+cu128, CUDA runtime 12.8, Braindecode 1.5.1, MNE 1.13.2, MOABB 1.7.2,
359 NumPy 2.5.3, SciPy 1.18.1, and scikit-learn 1.9.1. The continuation status records h5py 3.16.0,
360 joblib 1.6.0, pandas 3.0.6, skorch 1.4.0, matplotlib 3.11.2, and threadpoolctl 3.7.0. These are
361 recorded run versions, not a promise that later dependency resolution will produce identical
362 binaries. Historical Q5 and later development runtime changes remain visible in their receipts;
363 they limit contrasts that are not matched for runtime.

364 Migration restored source artifacts from the pinned GitHub snapshot and reconstituted external
365 epochs from checksum-verified R2 originals. The public restoration provenance records 160
366 verified external originals and 106 epoch-person artifacts. This should not be conflated with the
367 broader 178-file transport inventory, which additionally contains 18 BNCI source originals. R2 is
368 transport/storage infrastructure, not a fifth research dataset. The decoder-number audit recomputes
369 metrics from committed predictions without loading raw EEG or checkpoints. The separate Q16
370 analysis reads native BNCI raw EEG for spectral summaries. Both preserve historical fit counts
371 and perform zero decoder fits and zero new decoder-inference rows; external raw files are not
372 newly reprocessed by this manuscript revision.

373 **S9. Interpretation boundaries**

374 Development comparisons repeatedly use the same nine BNCI people and often reuse inner
375 trajectories, checkpoints, source subsets or endpoints. They describe sensitivity within this
376 development benchmark rather than independent confirmatory experiments. Q14 and Q15 have
377 different channel/window/reference contracts and are not unchanged-protocol replications. The
378 fixed frequency split and logit-average architecture establish neither physiological source
379 separation nor artifact causality, domain invariance, clinical benefit or online readiness. Calibrated
380 acquisition metadata, prespecified hardware harmonization, independent cohorts and causal
381 preprocessing are questions for future work, not completed analyses.

382 **S10. Descriptive BNCI-only physiological characterization (Q16)**

383 Q16-P001 executes the BNCI component of the earlier physiology proposal. Its explicit estimator,
384 aggregation, coverage, plotting and association specification was fixed on 6 October 2026 after
385 decoder outcomes were available and before the new raw-power calculation. It is descriptive, not
386 independent prospective registration or a new decoder experiment. The controlling protocol is
387 `research_runs/Q16-P001-BNCI-20261006/PROTOCOL.md`. External physiological analysis
388 remains unrun; BNCI results cannot establish an external neural mechanism or resolve the
389 acquisition limitations of Q15.

390 The declared population retains all 18 original BNCI MAT files, nine people, both sessions, 108
391 labelled motor-imagery runs and 5,184 four-class trials. The hand subset comprises 2,592 trials.
392 Provider flags remain annotations: the primary population includes the 488 flagged four-class and
393 246 flagged hand trials. Unflagged and session summaries are sensitivity descriptions rather than a
394 replacement population. All 18 originals matched their archived source SHA-256 and byte counts;
395 raw schema, labels, artifact flags, finite samples, all 5,184 event identities and complete windows
396 passed the metadata gate. All 228,096 trial-channel-band rows were eligible, with no nonfinite or
397 nonpositive power failure. No trial, person or required session was excluded.

398 Native sample rate is 250 Hz. MATLAB one-based trial starts are converted by subtracting one;
 399 the imagery cue is 500 samples later. Baseline cue $[-1.5, -0.5)$ s corresponds to native trial-start
 400 offsets $[125, 375)$, while task cue $[0.5, 2.5)$ s corresponds to $[625, 1125)$. Both windows must stay in
 401 the same native labelled run. Previous nominal imagery ends at cue +4 s; the metadata gate checks
 402 separation from that interval and treats first trials separately. Those checks establish nominal
 403 timing feasibility, not observed physiological recovery or independently recorded rest.
 404 Use the provider-ordered 22 EEG columns, excluding three EOG columns: Fz, FC3, FC1, FCz,
 405 FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P1, Pz, P2 and POz. This
 406 montage is documented from the provider, not supplied as individual MAT column-name strings.
 407 C3, Cz and C4 occupy zero-based columns 7, 9 and 11. Native signals receive no additional digital
 408 bandpass, notch, resampling, CAR, source scaler or learned projection. Provider acquisition
 409 filtering and reference remain part of the recordings. Numeric amplitudes are not newly
 410 authenticated physical microvolts; PSD has native-numeric squared per Hz units, integrated
 411 powers native-numeric squared units, and the task/baseline log ratio is dimensionless dB.
 412 Welch uses float64 samples; a one-second 250-sample periodic Hann taper (fftbins=True); 125-
 413 sample overlap; 250-point FFT; per-segment linear detrending; one-sided density scaling; and
 414 arithmetic segment averaging. The frequency grid is 1 Hz. Integrate inclusive 8–13-Hz mu and 13
 415 –30-Hz beta masks using trapezoidal integration in Hz. The common 13-Hz endpoint supplies
 416 half-bin weights to each adjacent band integral rather than a duplicated full-bin sum. Every
 417 trial/channel/band requires finite, strictly positive baseline and task power. Invalid pairs are
 418 reported explicitly; no arbitrary epsilon, clipping or outcome-based window changes are permitted.
 419 The trial metric is $10 \log_{10}(P_{\text{task}}/P_{\text{baseline}})$. Negative estimates mean lower task power relative
 420 to this window and estimator; positive estimates mean higher power. First average trial dB within
 421 person $\times$ session $\times$ class $\times$ channel $\times$ band, then average the two sessions equally within person,
 422 and then weight people equally. This averages log ratios and differs from the logarithm of mean
 423 powers. Preserve per-trial powers and validity, session means, person means and medians. Hand-

424 specific C3/C4 eligibility is paired. Four-class, flagged/unflagged and session summaries remain
 425 descriptive.

426 The baseline has one Welch segment and the task three overlapping segments, with different
 427 estimator variances. Because the logarithm is concave, this difference can produce a positive dB
 428 offset even when stationary power is unchanged; overlap also correlates the task segments. The
 429 ratio is therefore a fixed-window descriptive estimate, not an unbiased ERD estimator or a
 430 mechanistic test. The baseline was not redesigned after results were inspected.

431 The signed hand descriptor is $L = 0.5 \times [(C3_{\text{right}} - C4_{\text{right}}) + (C4_{\text{left}} - C3_{\text{left}})]$, calculated from
 432 the person-level dB means. A negative descriptor means relatively more negative contralateral
 433 than ipsilateral change; it can occur through ipsilateral increases without absolute contralateral
 434 suppression. Report the four constituent values, contralateral/ipsilateral means and Cz alongside it.
 435 Neither scalp topography nor this descriptor localizes a cortical source or proves that EEGNet uses
 436 the depicted signals. Relative ratios cancel only a constant multiplicative gain; they do not repair
 437 reference, timing, montage or artifact assumptions.

438 The primary contextual association uses each person's already saved three-seed mean balanced
 439 accuracy for the Q14-E001 matched-binary BNCI LOSO broad EEGNet. Shared EEGNet and CSP
 440 are retained as context, giving six Spearman coefficients across three fixed models and two fixed
 441 bands. Every person contributes 288 binary trials to that archived endpoint. All-nine-source fitted
 442 models and external outcomes are ineligible for this association. There are no physiological p-
 443 values, tertiles, high/low-accuracy groups, decoder correctness strata, favourable-person
 444 exclusions or model/band selection by correlation. Nine people remain nine person-level records
 445 despite seeds, sessions and trials, and overlapping LOSO source sets limit independence. The
 446 decoder uses cue [0.5,3.5) s while Q16 uses [0.5,2.5) s.

447 Scalp maps use a frozen schematic standard-1020 geometry for all 22 sensors, anterior upward and
 448 left scalp on viewer left, with linear triangulation restricted to the sensor convex hull. All
 449 hand/band maps share symmetric -6 to +6 dB colour limits; outside-hull regions remain blank and

450 actual clipping counts are reported. Template positions are not digitized participant anatomy.

451 Show C3/Cz/C4 person values, both-band laterality distributions and all nine broad-score

452 association points; shared/CSP contexts remain in tables or supplement. No separately unspecified

453 time-frequency method is introduced after outcomes are inspected.

454 A separate numerical review passed before manuscript insertion. The controlling parameter freeze

455 is commit 050e01b028aaab8e3d745934b13b2d17e9bb0a7a, published and read back before raw-

456 power execution on 6 October 2026. Its preprocessing_freeze.json SHA-256 is

457 c07a25e5de0c608cf405e728c0c85d1e0fae74e8e916d36e2cffad17d2ee3c01. The complete

458 execution run_manifest.json binds the scripts, native input receipt and 11 output artifacts; its

459 SHA-256 is 2fe8b8d1df751c440f35c7311691220668857d45ecac6734220fb12c8dc379d5. Exact

460 bindings are preserved with the freeze/readback, raw receipt, metadata audit, execution manifest

461 and independent_validation.json in evidence/q16_analysis/. This repository freeze follows known

462 decoder outcomes and is not independent preregistration.

463 Independent raw spectral replay covered every C3/Cz/C4 trial-band pair: 31,104 pairs and 62,208

464 baseline/task comparisons. An independently written closed-form linear detrend, explicit periodic

465 Hann taper, NumPy real FFT, one-sided periodogram scaling and manual trapezoid integration

466 reproduced the fixed estimator. Maximum relative power difference was $1.847193004695259e-15$;

467 maximum absolute dB difference was $7.105427357601002e-15$, within the fixed $1e-10$ tolerances.

468 This raw FFT replay covers the three central channels rather than all 22 spectra. All 228,096

469 output rows had their identities and eligibility checked, and every saved session, person, artifact-

470 stratum, laterality and six-coefficient association table was recomputed (aggregation absolute

471 tolerance $1e-11$). The saved-performance review rechecked 63 Q14 prediction files, 18,144

472 prediction rows and 135 source-only LOSO fit manifests. It loaded no model and generated no

473 prediction. This is computational validation of the fixed descriptive estimator, not a new external

474 experiment or independent laboratory replication.

475 The completed equal-person hand summaries are:

476 Methods table M1. Equal-person hand power summaries.

Band	Mean contralateral dB	Mean ipsilateral dB	Mean signed L dB	Person L range dB	Negative / positive people
Mu, 8–13 Hz	−0.308011	−0.058326	−0.249685	−1.432246 to +0.336434	6 / 3
Beta, 13–30 Hz	−0.439499	−0.271624	−0.167875	−0.519318 to +0.132853	8 / 1

477 Values repeat the independently checked Q16 results; no additional analysis is performed.

478 Both group absolute components were negative, but this was not true for every person. S2

479 illustrates the distinction: mu L was −0.202912 dB while contralateral and ipsilateral changes were

480 +1.490894 and +1.693806 dB. Session, hand and flagged/unflagged cells remain available in the

481 independently checked tables; these records do not establish artifact-free physiology or replace the

482 primary all-trial population. The scalp-map export receipt records zero sensor and interpolated-

483 grid values clipped by the fixed ± 6 -dB scale in each of the four hand/band maps.

484 All six signed physiology–accuracy Spearman coefficients are reported without p-values:

485 Methods table M2. All descriptive physiology–accuracy coefficients.

Saved Q14-E001 binary LOSO model	Mu rho	Beta rho	People
Broad EEGNet, designated descriptive endpoint	−0.150000	+0.600000	9
Shared mu/beta EEGNet, context	−0.083333	+0.483333	9
CSP4+LDA, context	+0.066667	+0.316667	9

486 Values repeat the independently checked Q16 results; no additional analysis is performed.

487 Positive beta rho associates higher balanced accuracy with a larger, less-negative signed laterality

488 descriptor. It therefore does not indicate that stronger relative contralateral suppression improved

489 balanced accuracy. The same nine people appear in all six panels, without exclusions, fitted lines,
490 p-values or subgroup thresholds. Figures use the independently verified table values, with
491 input/output hashes in evidence/figures_q16.json.

492 This signal analysis adds zero decoder fits and zero checkpoint-inference rows, and does not
493 modify archived decoder checkpoints, predictions, statistical plans or tests. External physiology
494 was not computed. The nine-person associations remain post-decoder-outcome descriptive context
495 and do not localize cortical sources, establish decoder attribution or validate a biomarker.

496 **References**

- 497 1. Brunner C, Leeb R, Müller-Putz GR, Schlögl A, Pfurtscheller G. BCI Competition 2008 – Graz
498 data set A. Official competition dataset description. 2008. Available from:
499 https://www.bbc.de/competition/iv/desc_2a.pdf [cited 2026 Oct 8].
- 500 2. Tangermann M, Müller KR, Aertsen A, Birbaumer N, Braun C, Brunner C, et al. Review of the
501 BCI Competition IV. *Front Neurosci.* 2012;6. doi:10.3389/fnins.2012.00055
- 502 3. Jayaram V, Barachant A. MOABB: trustworthy algorithm benchmarking for BCIs. *J Neural*
503 *Eng.* 2018;15(6):066011. doi:10.1088/1741-2552/aadea0
- 504 4. Lotte F, Bougrain L, Cichocki A, Clerc M, Congedo M, Rakotomamonjy A, et al. A review of
505 classification algorithms for EEG-based brain–computer interfaces: a 10 year update. *J Neural*
506 *Eng.* 2018;15(3):031005. doi:10.1088/1741-2552/aab2f2
- 507 5. Schalk G, McFarland DJ, Hinterberger T, Birbaumer N, Wolpaw JR. EEG Motor
508 Movement/Imagery Dataset. dataset. 2009. doi:10.13026/C28G6P
- 509 6. Schalk G, McFarland DJ, Hinterberger T, Birbaumer N, Wolpaw JR. BCI2000: A General-
510 Purpose Brain-Computer Interface (BCI) System. *IEEE Trans Biomed Eng.* 2004;51(6):1034-
511 1043. doi:10.1109/TBME.2004.827072
- 512 7. Cho H, Ahn M, Ahn S, Kwon M, Jun SC. EEG datasets for motor imagery brain–computer
513 interface. *GigaScience.* 2017;6(7). doi:10.1093/gigascience/gix034

514 8. Cho H, Ahn M, Ahn S, Kwon M, Jun SC. Supporting data for "EEG datasets for motor imagery
515 brain computer interface". dataset. 2017. doi:10.5524/100295

516 9. Lee MH, Kwon OY, Kim YJ, Kim HK, Lee YE, Williamson J, et al. EEG dataset and OpenBMI
517 toolbox for three BCI paradigms: an investigation into BCI illiteracy. GigaScience. 2019;8(5).
518 doi:10.1093/gigascience/giz002

519 10. Lee MH, Kwon OY, Kim YJ, Kim HK, Lee YE, Williamson J, et al. Supporting data for "EEG
520 Dataset and OpenBMI Toolbox for Three BCI Paradigms: An Investigation into BCI Illiteracy".
521 dataset. 2019. doi:10.5524/100542

522 11. Gulrajani I, Lopez-Paz D. In Search of Lost Domain Generalization. International Conference
523 on Learning Representations. 2021. Available from: <https://iclr.cc/virtual/2021/poster/2998> [cited
524 2026 Oct 8].

525 12. Ramoser H, Muller-Gerking J, Pfurtscheller G. Optimal spatial filtering of single trial EEG
526 during imagined hand movement. IEEE Trans Rehabil Eng. 2000;8(4):441-446.
527 doi:10.1109/86.895946

528 13. Ang KK, Chin ZY, Wang C, Guan C, Zhang H. Filter Bank Common Spatial Pattern
529 Algorithm on BCI Competition IV Datasets 2a and 2b. Front Neurosci. 2012;6.
530 doi:10.3389/fnins.2012.00039

531 14. Ang KK, Chin ZY, Zhang H, Guan C. Filter Bank Common Spatial Pattern (FBCSP) in Brain-
532 Computer Interface. 2008 IEEE International Joint Conference on Neural Networks (IEEE World
533 Congress on Computational Intelligence). 2008:2390-2397. doi:10.1109/IJCNN.2008.4634130

534 15. Lawhern VJ, Solon AJ, Waytowich NR, Gordon SM, Hung CP, Lance BJ. EEGNet: a compact
535 convolutional neural network for EEG-based brain-computer interfaces. J Neural Eng.
536 2018;15(5):056013. doi:10.1088/1741-2552/aace8c

537 16. Schirrmeister RT, Springenberg JT, Fiederer LDJ, Glasstetter M, Eggersperger K,
538 Tangermann M, et al. Deep learning with convolutional neural networks for EEG decoding and
539 visualization. Hum Brain Mapp. 2017;38(11):5391-5420. doi:10.1002/hbm.23730

- 540 17. Zhong XC, Wang Q, Liu D, Chen Z, Liao JX, Sun J, et al. EEG-DG: A Multi-Source Domain
541 Generalization Framework for Motor Imagery EEG Classification. *IEEE J Biomed Health Inform.*
542 2025;29(4):2484-2495. doi:10.1109/jbhi.2024.3431230
- 543 18. Zheng Y, Wu S, Chen J, Yao Q, Zheng S. Cross-Subject Motor Imagery
544 Electroencephalogram Decoding with Domain Generalization. *Bioengineering (Basel).*
545 2025;12(5):495. doi:10.3390/bioengineering12050495
- 546 19. Jiang WB, Zhao LM, Lu BL. Large Brain Model for Learning Generic Representations with
547 Tremendous EEG Data in BCI. *The Twelfth International Conference on Learning*
548 *Representations*. 2024. Available from: <https://openreview.net/forum?id=QzTpTRVtrP> [cited 2026
549 Oct 8].
- 550 20. Wang G, Liu W, He Y, Xu C, Ma L, Li H. EEGPT: Pretrained Transformer for Universal and
551 Reliable Representation of EEG Signals. *Advances in Neural Information Processing Systems*.
552 2024;37. doi:10.52202/079017-1239
- 553 21. Pfurtscheller G, Lopes da Silva FH. Event-related EEG/MEG synchronization and
554 desynchronization: basic principles. *Clin Neurophysiol.* 1999;110(11):1842-1857.
555 doi:10.1016/s1388-2457(99)00141-8
- 556 22. Pfurtscheller G, Neuper C. Motor imagery activates primary sensorimotor area in humans.
557 *Neurosci Lett.* 1997;239(2-3):65-68. doi:10.1016/s0304-3940(97)00889-6
- 558 23. Sagawa S, Koh PW, Hashimoto TB, Liang P. Distributionally Robust Neural Networks for
559 Group Shifts: On the Importance of Regularization for Worst-Case Generalization. *International*
560 *Conference on Learning Representations*. 2020. Available from: [https://openreview.net/forum?](https://openreview.net/forum?id=ryxGuJrFvS)
561 [id=ryxGuJrFvS](https://openreview.net/forum?id=ryxGuJrFvS) [cited 2026 Oct 8].
- 562 24. Efron B. Bootstrap Methods: Another Look at the Jackknife. *Ann Stat.* 1979;7(1).
563 doi:10.1214/aos/1176344552
- 564 25. Winkler AM, Ridgway GR, Webster MA, Smith SM, Nichols TE. Permutation inference for
565 the general linear model. *Neuroimage.* 2014;92:381-397. doi:10.1016/j.neuroimage.2014.01.060

- 566 26. Holm S. A Simple Sequentially Rejective Multiple Test Procedure. *Scand J Stat.* 1979;6(2):65
567 –70. Available from: <https://www.jstor.org/stable/4615733> [cited 2026 Oct 8].
- 568 27. Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PC, Mark RG, et al.
569 PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for
570 Complex Physiologic Signals. *Circulation.* 2000;101(23). doi:10.1161/01.CIR.101.23.e215
- 571 28. Pollard T, Moody BE, Lehman LwH, Gow BJ, Fernandes C, Xie C, et al. PhysioNet as a
572 global platform for biomedical research. *Nat Health.* 2026;1(8):792-795. doi:10.1038/s44360-026-
573 00096-z